MILITARY SPECIFICATION SHEET

# CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, 45°, SELF-LOCKING AND NONSELF-LOCKING, CATEGORY 4C (FOR MIL-C-38999 SERIES I AND II CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the accessories described herein shall consist of this specification sheet and the latest issue of MIL-C-85049.

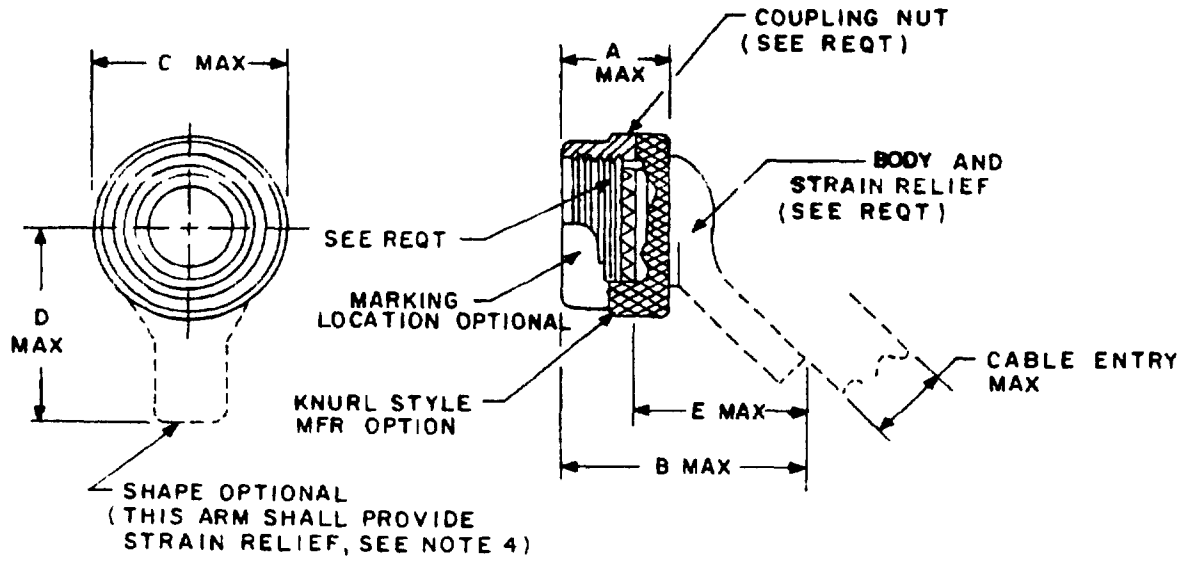



FIGURE 1. Dimensions and configurations (self-locking and nonself-locking).


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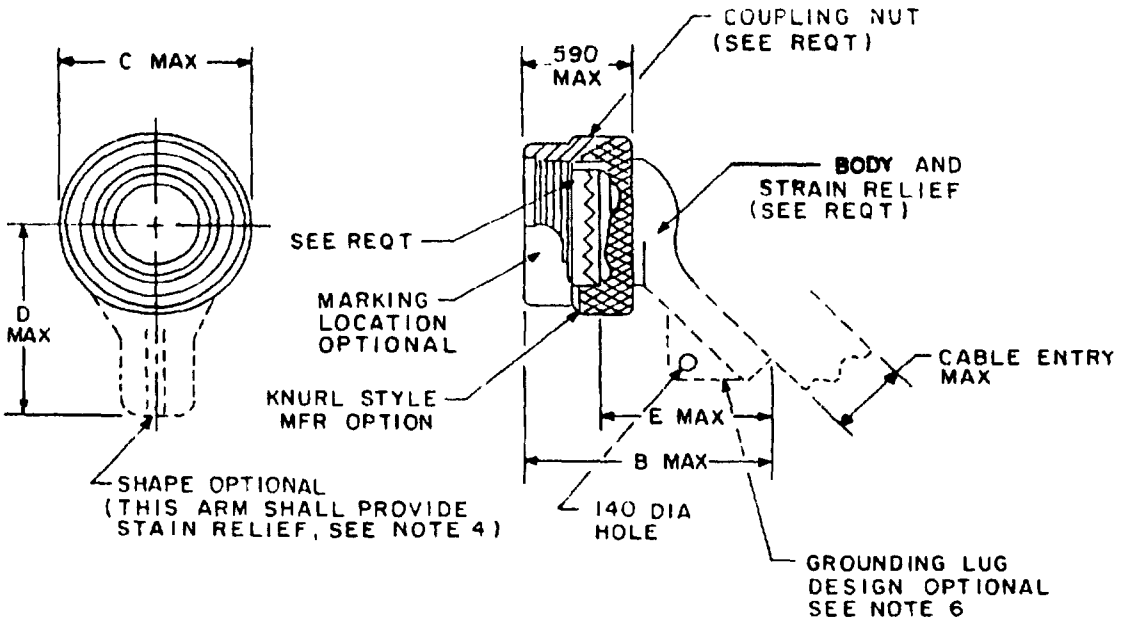
Shell sizes and dimensions

| Dash number | Connector shell size | | A max | | B max | C max | | D max | E max | Cable entry max |
|---|---|---|---|---|---|---|---|---|---|---|
| | Series II | Series I | Non-self-locking | Self-locking | | Non-self-locking | Self-locking | | | |
| 8 | 8 | 9 | .540 (13.7) | .590 (14.99) | 1.25 (31.8) | .75 (19.1) | .86 (21.8) | .88 (22.3) | 1.03 (26.2) | .264 (6.71) |
| 10 | 10 | 11 | .540 (13.7) | .590 (14.99) | 1.25 (31.8) | .85 (21.6) | .98 (24.9) | .88 (22.3) | 1.03 (26.2) | .392 (9.96) |
| 12 | 12 | 13 | .540 (13.7) | .590 (14.99) | 1.25 (31.8) | 1.00 (25.4) | 1.16 (29.5) | .94 (23.9) | 1.03 (26.2) | .506 (12.9) |
| 14 | 14 | 15 | .540 (13.7) | .590 (14.99) | 1.31 (33.3) | 1.10 (27.9) | 1.28 (32.5) | 1.25 (31.8) | 1.09 (27.7) | .631 (16.0) |
| 16 | 16 | 17 | .540 (13.7) | .590 (14.99) | 1.33 (33.8) | 1.25 (31.8) | 1.41 (35.8) | 1.31 (33.3) | 1.11 (28.2) | .756 (19.2) |
| 18 | 18 | 19 | .540 (13.7) | .590 (14.99) | 1.43 (36.3) | 1.40 (35.6) | 1.52 (38.6) | 1.38 (35.1) | 1.21 (30.7) | .845 (21.5) |
| 20 | 20 | 21 | .540 (13.7) | .590 (14.99) | 1.48 (37.6) | 1.50 (38.1) | 1.64 (41.7) | 1.50 (38.1) | 1.26 (32.0) | .970 (24.6) |
| 22 | 22 | 23 | .540 (13.7) | .590 (14.99) | 1.52 (38.6) | 1.65 (41.9) | 1.77 (45.0) | 1.69 (42.9) | 1.30 (33.0) | 1.095 (27.81) |
| 24 | 24 | 25 | .540 (13.7) | .590 (14.99) | 1.56 (39.6) | 1.75 (44.5) | 1.89 (48.0) | 1.75 (44.5) | 1.34 (34.0) | 1.220 (30.99) |

NOTES:
1. Dimensions are in inches.
2. Tolerances shall be: .xx = ±.03 and .xxx = ±0.15 angular tolerances x = ±2.
3. Metric equivalents are in parentheses and are given for general information only.
4. Strain relief unit is used with MS3367-1-X plastic strap, MIL-T-713 lacing twine or MIL-T-43435 lacing tape or equivalent, not supplied with the backshell.
5. This accessory will not accommodate connectors using 8, 4, and 0 contacts.

FIGURE 1. Dimensions and configurations (self-locking and nonself-locking) - Continued.

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Shell sizes and dimensions

| Dash number | Connector Shell size | | B max | C dia max | D max | E max | Cable entry | Dash number | Connector Shell size | | B max | C dia max | D max | E max | Cable entry max |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Series II | Series I | | | | | | | Series II | Series I | | | | | |
| 8 | 8 | 9 | 1.40 (35.6) | .859 (21.82) | .88 (22.3) | 1.21 (30.7) | .264 (6.71) | 16 | 16 | 17 | 1.58 (40.1) | 1.406 (35.71) | 1 31 (33.3) | 1.39 (35.3) | .756 (19.20) |
| 10 | 10 | 11 | 1.40 (35.6) | .984 (24.99) | .88 (22.3) | 1.21 (30.7) | .392 (9.96) | 18 | 18 | 19 | 1.58 (40.1) | 1 516 (38.51) | 1.38 (35.1) | 1.39 (35.3) | .845 (21 46) |
| 12 | 12 | 13 | 1.58 (40.1) | 1.156 (29.36) | .94 (23.8) | 1.39 (35.3) | .506 (12.85) | 20 | 20 | 21 | 1.58 (40.1) | 1.641 (41.68) | 1.50 (38.1) | 1.39 (35.3) | .970 (24.64) |
| 14 | 14 | 15 | 1.58 (40.1) | 1.281 (32.54) | 1.25 (31.8) | 1.39 (35.3) | .631 (16.03) | 22 | 22 | 23 | 1.58 (40.1) | 1.766 (44.86) | 1.69 (42.9) | 1.39 (35.3) | 1.095 (27.81) |
| | | | | | | | | 24 | 24 | 25 | 1.58 (40.1) | 1.891 (48.03) | 1.75 (44 5) | 1.39 (35 3) | 1.220 (30 99) |

NOTES:
1. Dimensions are in inches.
2. Tolerances shall be .xx = ±.03 and .xxx = ±.015 angular tolerances x = ±2.
3. Metric equivalents are in parentheses and are given for general information only.
4. Strain relief unit is used with MS3367-1-X plastic strap, MIL-T-713 lacing twine or MIL-T-43435 lacing tape or equivalent, not supplied with the backshell.
5. This accessory will not accommodate connectors using 8, 4, and 0 contacts.
6. Not for Navy use. Not for new design for Air Force.

FIGURE 2. Dimensions and configurations (self-locking only with grounding lug).



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REQUIREMENTS

Self-locking and nonself-locking styles

Design and construction

Dimensions and configurations  See figures 1 and 2.

Interface dimensions  In accordance with MIL-C-38999, series I and II.

Accessory  Consists of a coupling nut and body with strain relief arm. The coupling nut shall be captivated to the body and free to rotate.

Material

Coupling nut and follower.  Aluminum alloy in accordance with MIL-C-85049.

Tie mount.  Polyamide (optional).

Fastener  Stainless steel (optional).

Finish.  In accordance with MIL-C-85049, A, N or W.  1/

Self-locking styles only

Self-locking device within the coupling nut shall be corrosion-resistant metal and shall provide a positive detent.  Couplings with self-locking devices shall meet all the performance requirements specified herein for its specific category and the following additional tests.

Life cycle.  The coupling with the locking device engaged shall be rotated 60 full turns in a clockwise direction and 60 full turns counter clockwise, prior to standard qualification testing.  Cycle speed shall not exceed one full turn per second.

Locking:  Coupling nut torque shall be 80 percent of coupling thread strength values given for medium and light duty accessories as specified in table III of MIL-C-85049.  In table III where N/A appears, the values shall be 80 percent of those values used in qualification testing for coupling thread strength.  Tolerance shall be ±5 inch pounds.

Vibration  Backshells shall be subjected to test parameters of applicable connector specifications.  The coupling torque shall be within +20, -10 inch-pounds of the initial value.

Part number  See example below

| M85049/ | 57 | S | 10 | W |
|---|---|---|---|---|

Military designation: M85049/

Specification sheet: 57

Configuration: S - for self-locking, G - for self-locking with grounding lug, use a dash for nonself locking: S

Dash number (see figures 1 and 2): 10

Finish: W

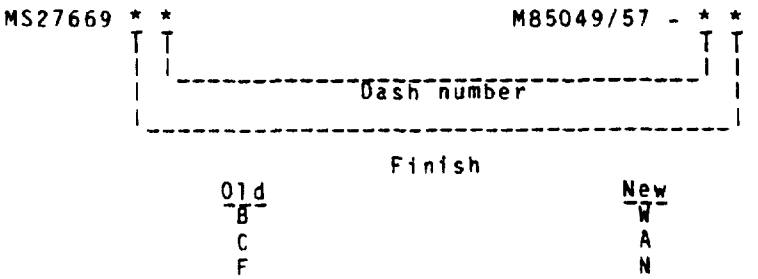
1/ A finish inactive for new design.  N finish not for Navy use, Air Force use for space applications only.  W is the prefered finish.

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Qualification. See MIL-C-85049, category 4C, cable pullout not applicable. Accessory shall be tested with MIL-C-38999, series I connectors.

QPL evaluating activity: Naval Avionics Center, Code B/714, 6000 East 21st Street, Indianapolis, IN 46219-2189.

Supersession data:




Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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